

Deploy and Connect to a Windows Server 2025 Virtual Machine in Microsoft Azure

Summary:-

In this lab, a Windows Server 2025 virtual machine is deployed in Microsoft Azure using the Azure Virtual Machine service. The virtual machine is configured with the required compute, networking, storage, and security settings. After deployment, the associated Azure resources are reviewed, and a Remote Desktop Protocol (RDP) connection is established to access and verify the Windows Server virtual machine.

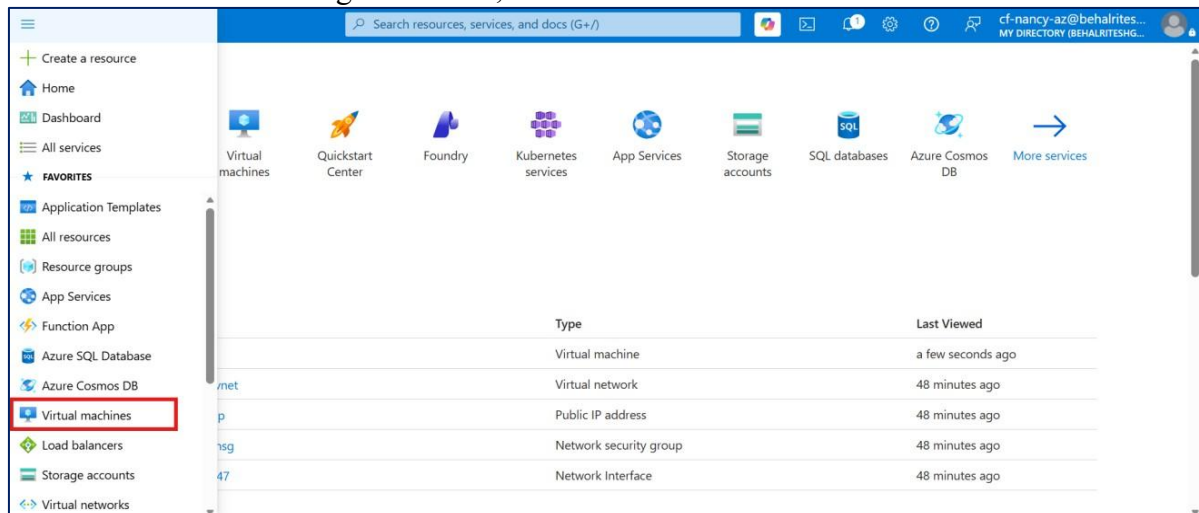
Let's Start:-

- Create a Resource Group.
- Deploy a Windows Server 2025 virtual machine.
- Configure compute, networking, storage, and security settings.
- Review the Azure resources created during deployment.
- Download the Remote Desktop (RDP) connection file.
- Connect to the virtual machine using Remote Desktop.
- Verify successful access to the Windows Server desktop.

Lab Steps:-

Step 1 – Open the Virtual Machines Service

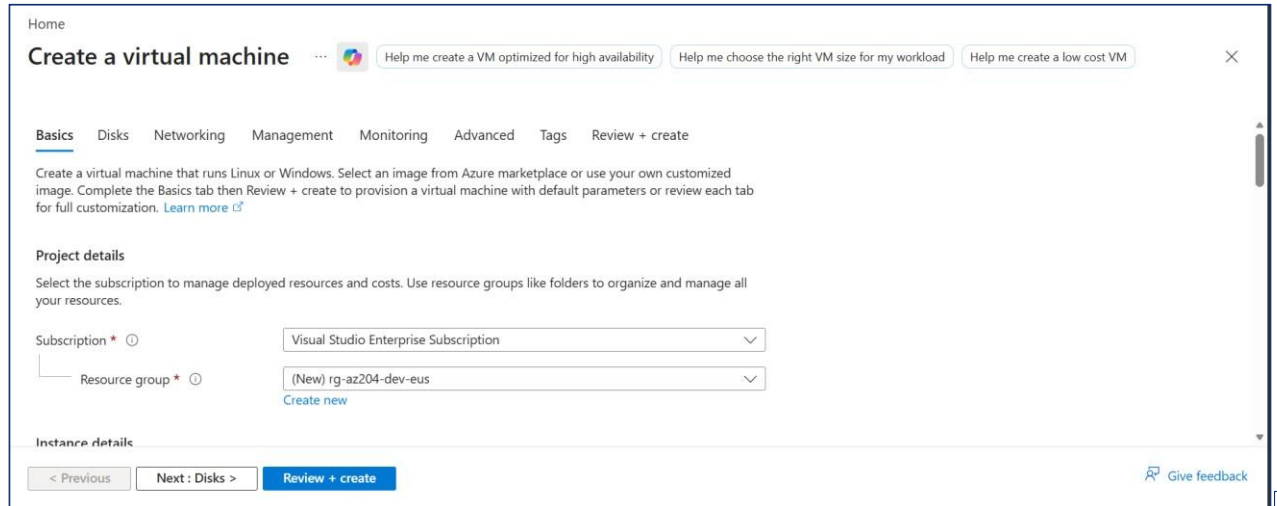
- Sign in to the **Azure Portal**.
- From the left navigation menu, select **Virtual Machines**.



- Click **Create** → **Azure Virtual Machine**.
- This step opens the Virtual Machine deployment wizard.

Step 2 – Configure the Basic Settings

- Configure the following:
- **Subscription:** Visual Studio Enterprise Subscription
- **Resource Group:** rg-az204-dev-eus (Create New)



Home

Create a virtual machine

Help me create a VM optimized for high availability | Help me choose the right VM size for my workload | Help me create a low cost VM

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | Review + create

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

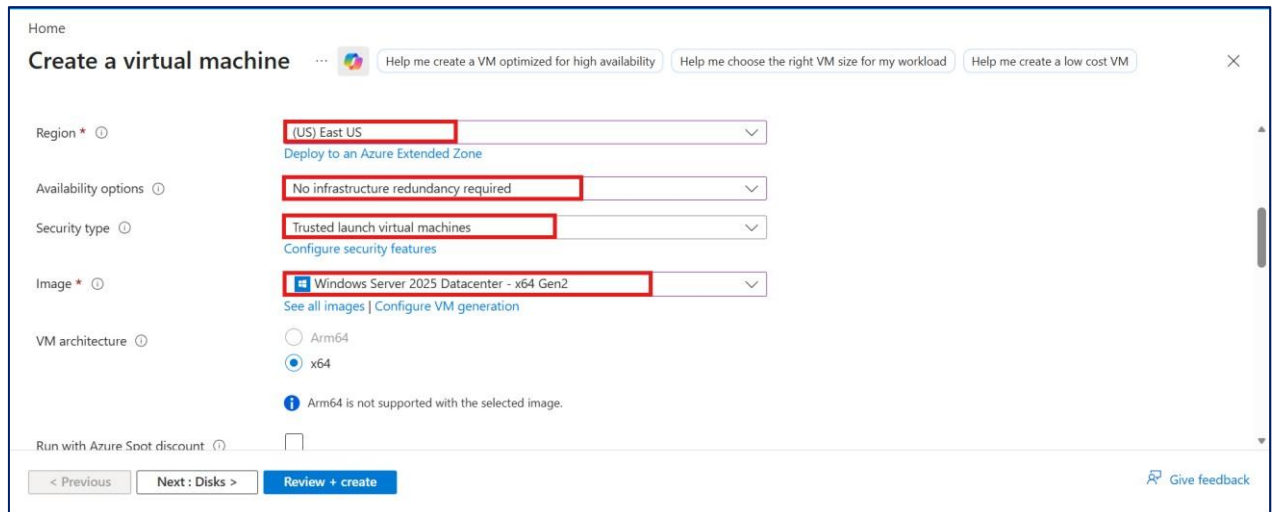
Resource group * [Create new](#)

Instance details

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- **Region:** (US) East US
- **Availability Options:** No infrastructure redundancy required
- **Security Type:** Trusted launch virtual machines
- **Image:** Windows Server 2025 Datacenter - x64 Gen2
- **VM Architecture:** x64



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Region * [Deploy to an Azure Extended Zone](#)

Availability options

Security type [Configure security features](#)

Image * [See all images](#) | [Configure VM generation](#)

VM architecture ☒ x64 ☐ Arm64

Arm64 is not supported with the selected image.

Run with Azure Spot discount ☐

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Select a VM size

Search by VM size... vCPUs: All RAM (GiB): All Display cost: Monthly Add filter

Showing 1356 VM sizes. Subscription: Visual Studio Enterprise Subscription Region: East US Current size: Standard_D2ds_v7 Image: Windows Server 2025 Datacenter Learn more about VM sizes Group by series

VM Size	Type	vCPUs	RAM (GiB)	Data disks	Max IOPS	Local storage (GiB)
D2ads_v7	General purpose	2	8	10	4000	110 (NVMe)
D2allds_v7	General purpose	2	4	10	4000	110 (NVMe)
D2als_v7	General purpose	2	4	10	4000	N/A
D2as_v7	General purpose	2	8	10	4000	N/A
D2ds_v7	General purpose	2 (Customize)	8	10	4000	110 (NVMe)

Select Prices presented are estimates in INR that include only Azure infrastructure costs and any discounts for the subscription and location. The prices don't include any applicable software costs. Final charges will appear in your local currency in cost analysis and billing views. View Azure pricing calculator. Give feedback

- Click **Select**.
- Continue configuring:
- **Administrator Username:** Enter your username • **Password:** Enter a strong password
- **Confirm Password:** Re-enter the password

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⚠ Changing Basic options may reset selections you have made. Review all options prior to creating the virtual machine.

Administrator account

Username * cf-nancy-az ✓

Password * ***** ✓

Confirm password * ***** ✓

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ☐ None ☒ Allow selected ports

< Previous Next: Disks > Review + create Give feedback

- **Public Inbound Ports:** Allow selected ports
- **Inbound Ports:** RDP (3389)
- Click **Next: Disks**.

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Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports *

☐ None

☒ Allow selected ports

Select inbound ports *

RDP (3389)

Warning: This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

Licensing

< Previous **Next : Disks >** Review + create

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- This step configures the virtual machine's basic deployment settings.

Step 3 – Configure the OS Disk

- **Configure** the following:
 OS Disk Size: Image default (127 GiB)
 OS Disk Type: Premium SSD (Locally Redundant Storage)
 Key Management: Platform-managed key
- Leave the remaining settings as **default**.
- Click **Next: Networking**.

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OS disk

OS disk size

Image default (127 GiB)

OS disk type *

Premium SSD (locally-redundant storage)

Delete with VM

☒

Key management

Platform-managed key

Enable Ultra Disk compatibility

☐

Ultra disk is supported in Availability Zone(s) 1,2,3 for the selected VM size Standard_D2ds_v7.

Data disks for vm-dev-eus-01

You can add and configure additional data disks for your virtual machine or attach existing disks. This VM also comes with a temporary disk.

< Previous **Next : Networking >** Review + create

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- This step configures the operating system storage for the virtual machine.

Step 4 – Configure Networking

- **Verify** the automatically created networking resources:
 Virtual Network: vm-dev-eus-01-vnet
 Subnet: default (10.0.0.0/24)

Public IP: vm-dev-eus-01-ip

- Leave the remaining settings unchanged.
- Click Next: **Management**.

The screenshot shows the 'Networking' tab of the 'Create a virtual machine' wizard. The 'Virtual network' is set to '(new) vm-dev-eus-01-vnet', the 'Subnet' is '(new) default (10.0.0.0/24)', and the 'Public IP' is '(new) vm-dev-eus-01-ip'. All three dropdowns are highlighted with red boxes. Below the dropdowns, there is a link to 'Create new' and a note about public IP charges. At the bottom, the 'Next: Management >' button is highlighted with a red box.

- This step configures network connectivity for the virtual machine.

Step 5 – Configure Management

- Leave the default management settings.
- Keep Microsoft Defender for Cloud enabled.
- Click Next: **Monitoring**.

The screenshot shows the 'Management' tab of the 'Create a virtual machine' wizard. The 'Microsoft Defender for Cloud' section is visible, with the 'Enable basic plan for free' checkbox checked. Below it, the 'Metadata Security Protocol' section is also visible. At the bottom, the 'Next: Monitoring >' button is highlighted with a red box.

- This step configures the default management options for the virtual machine.

Step 6 – Configure Monitoring

- Verify the following settings:
Recommended Alert Rules: Disabled
Boot Diagnostics: Enable with managed storage account

- Click Next: Advanced.

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Basics Disks Networking Management **Monitoring** Advanced Tags Review + create

Configure monitoring options for your VM.

Alerts

Enable recommended alert rules ⓘ ☐

Diagnostics

Boot diagnostics ⓘ

☒ Enable with managed storage account (recommended)

☐ Enable with custom storage account

☐ Disable

i There is a nominal charge for the managed storage account.

Enable OS guest diagnostics ⓘ ☐

< Previous **Next : Advanced >** Review + create [Give feedback](#)

- This step configures monitoring and diagnostic settings for the virtual machine.

Step 7 – Configure Advanced Settings

- Leave all advanced settings unchanged.
- Click Next: **Tags**.

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Basics Disks Networking Management Monitoring **Advanced** Tags Review + create

Add additional configuration, agents, scripts or applications via virtual machine extensions or cloud-init.

Extensions

Extensions provide post-deployment configuration and automation.

Extensions ⓘ [Select an extension to install](#)

VM applications

VM applications contain application files that are securely and reliably downloaded on your VM after deployment. In addition to the application files, an install and uninstall script are included in the application. You can easily add or remove applications on your VM after create. [Learn more](#) [↗](#)

[Select a VM application to install](#)

< Previous **Next : Tags >** Review + create [Give feedback](#)

- This step retains the default advanced configuration.

Step 8 – Configure Tags

- Do not add any tags.
- Click Next: **Review + Create**.

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Basics Disks Networking Management Monitoring Advanced **Tags** Review + create

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. [Learn more about tags](#)

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

Name	Value	Resource
	:	14 selected

< Previous **Next : Review + create >** Review + create

[Give feedback](#)

- This step skips optional resource tagging.

Step 9 – Review and Create

- Wait for validation to complete.
- Verify that Validation Passed is displayed.
- Review the deployment configuration.
- Click **Create**.

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Basics Disks Networking Management Monitoring Advanced Tags **Review + create**

Validation passed

Price

1 X Standard D2ds v7
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Subscription credits apply ⓘ
21.2146 INR/hr
[Pricing for other VM sizes](#)

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same

< Previous Next > **Create**

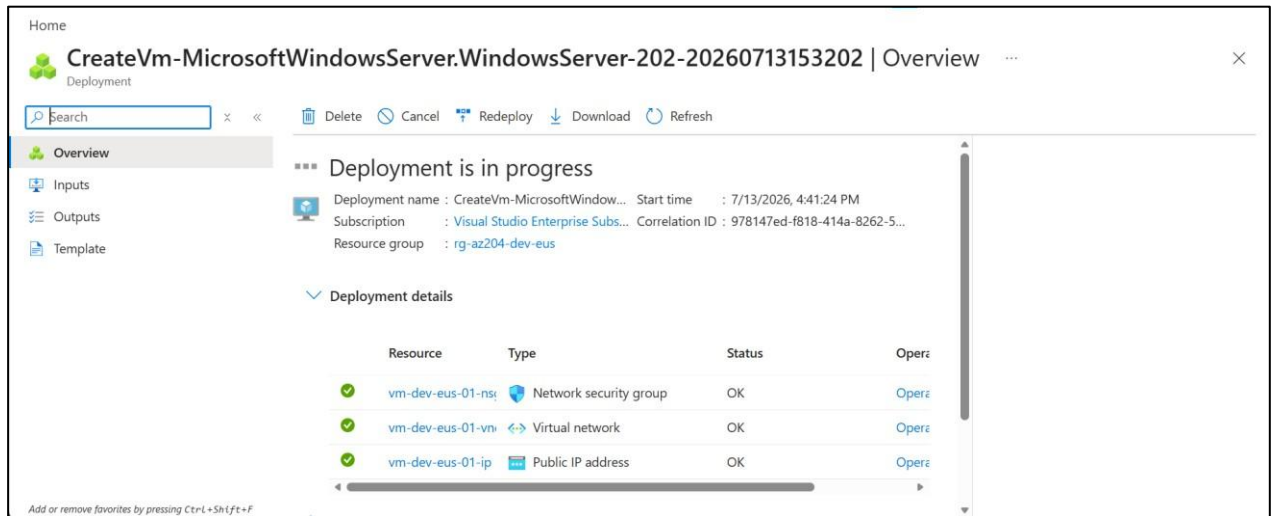
[Download a template for automation](#) [Give feedback](#)

- This step validates and starts the virtual machine deployment.

Step 10 – Verify the Deployment

- Wait until the deployment status changes to Deployment Complete.
- Verify that Azure has created the following resources:
 - Virtual Machine
 - Virtual Network
 - Subnet

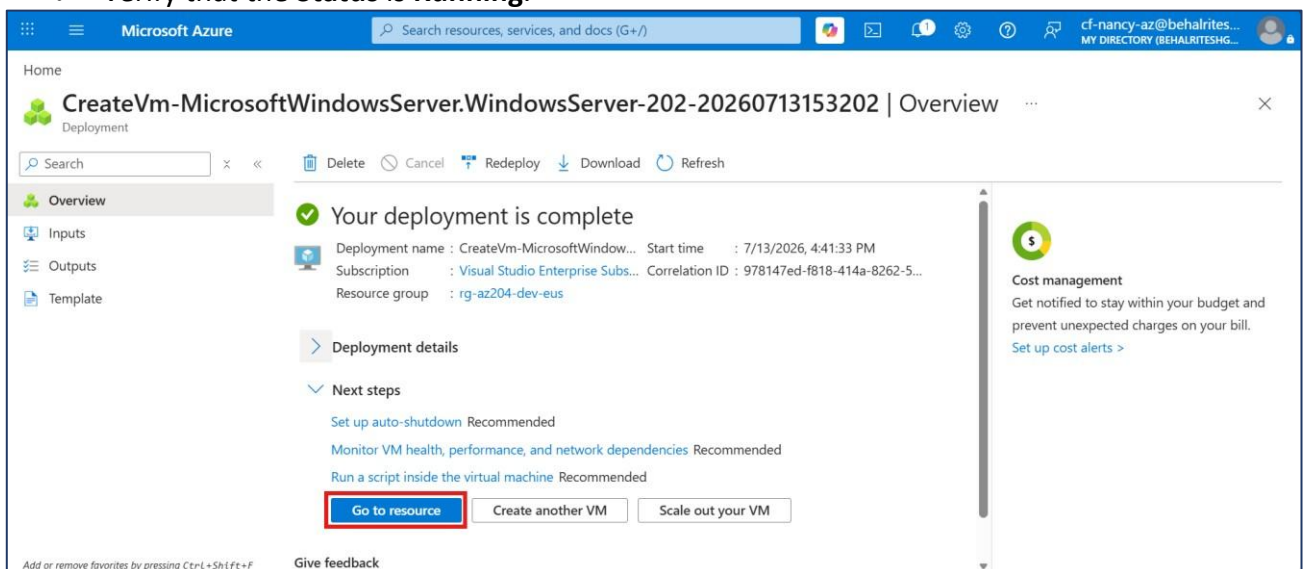
- o Public IP Address
- o Network Interface
- o Network Security Group
- o OS Disk



- This step confirms that the virtual machine and its associated resources have been deployed successfully.

Step 11 – Verify the Virtual Machine Deployment

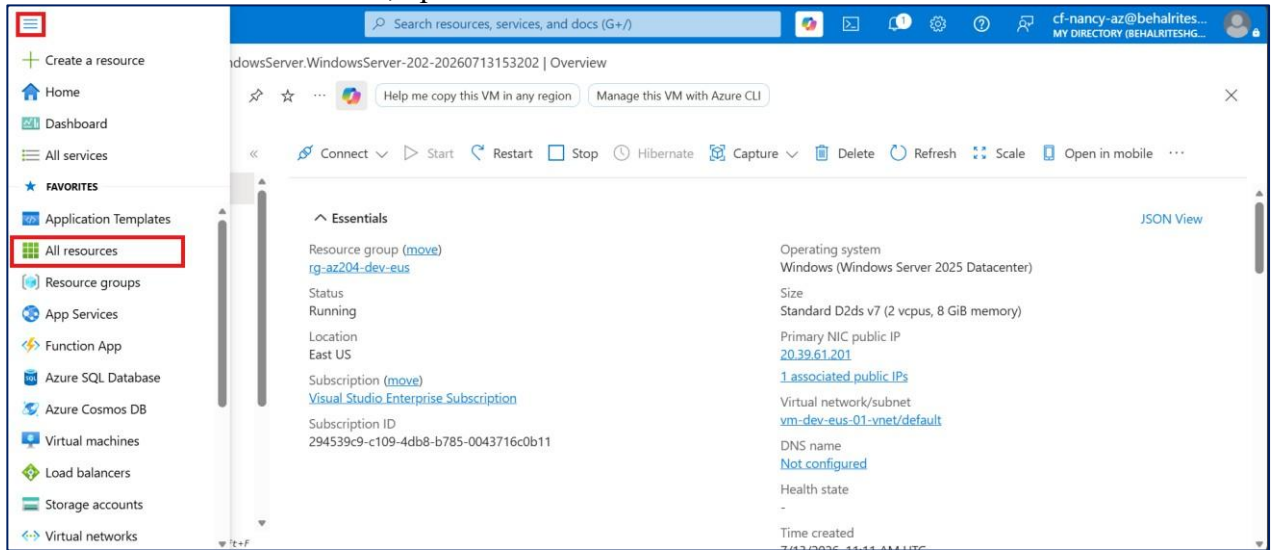
- Sign in to the **Azure Portal**.
- Open **Virtual Machines**.
- Select the previously created virtual machine.
- Verify that the **Status** is **Running**.



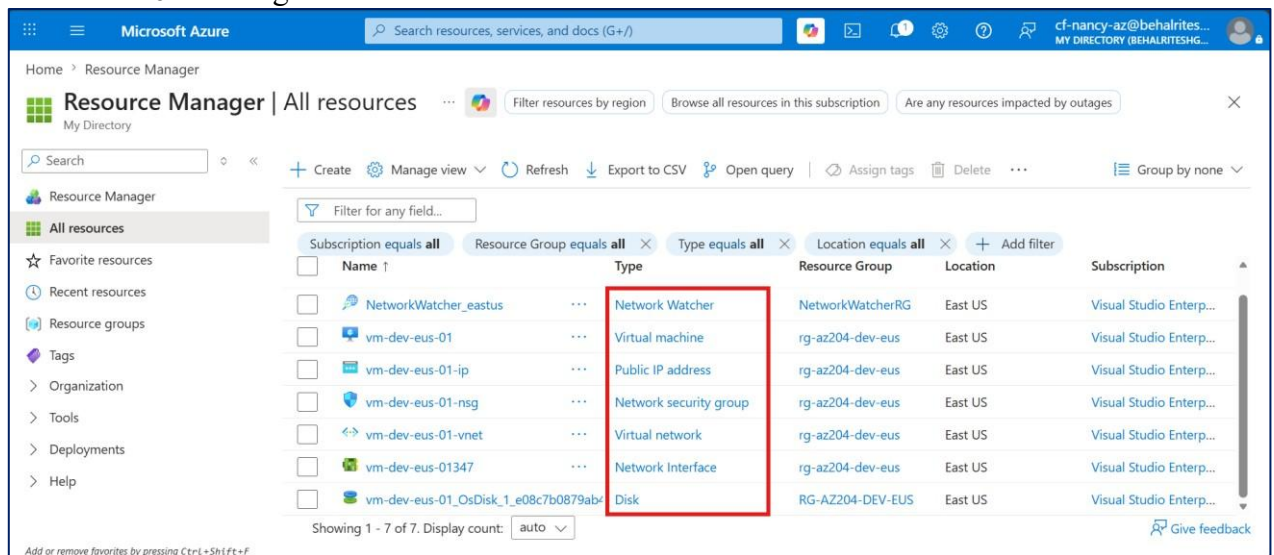
- This step verifies that the virtual machine is ready for remote access.

Step 12 – Review the Azure Resources

- From the Azure Portal, open **All Resources**



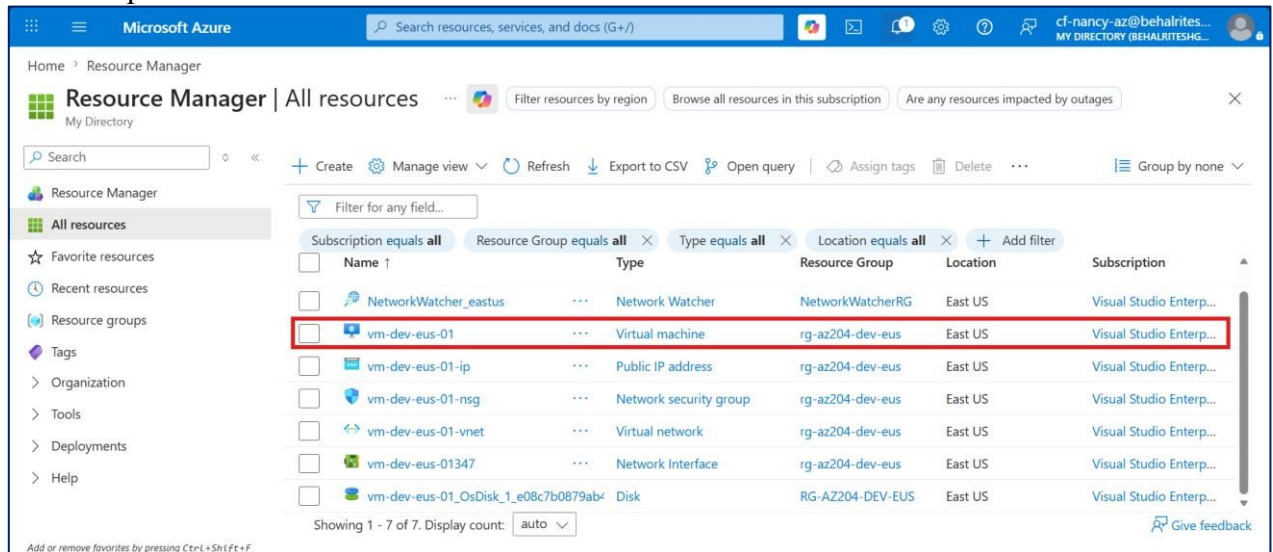
- Verify that the following resources were created:
 - Virtual Machine
 - Virtual Network
 - Public IP Address
 - Network Interface
 - Network Security Group
 - Managed Disk



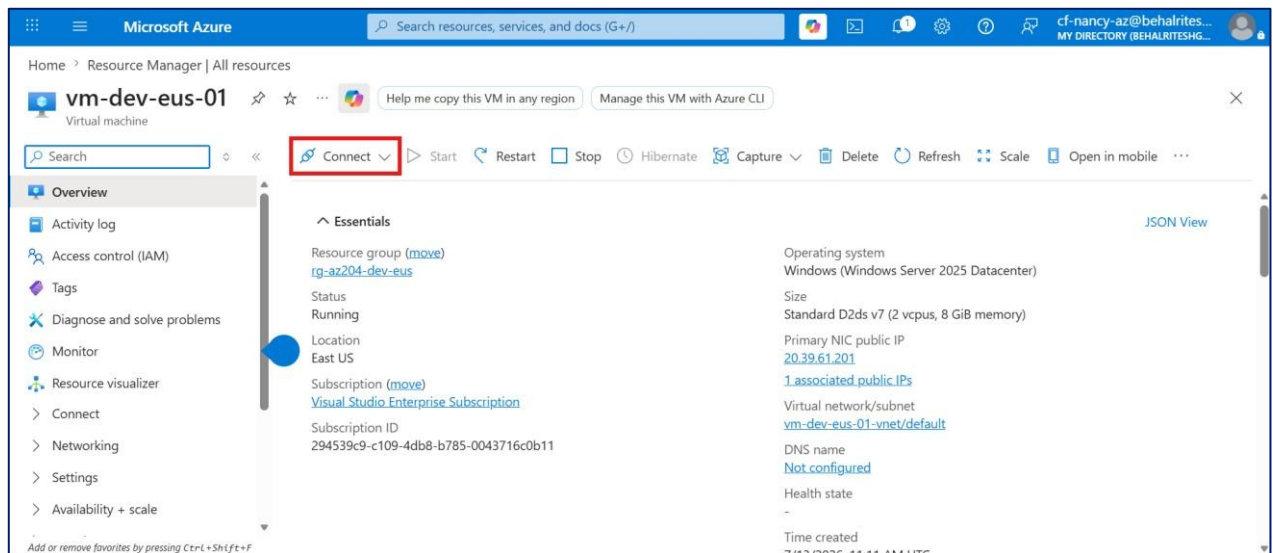
- Review the **Overview** page of the virtual machine.
- This step reviews the Azure resources automatically created during deployment.

Step 13 – Download the RDP File

- Open the virtual machine.

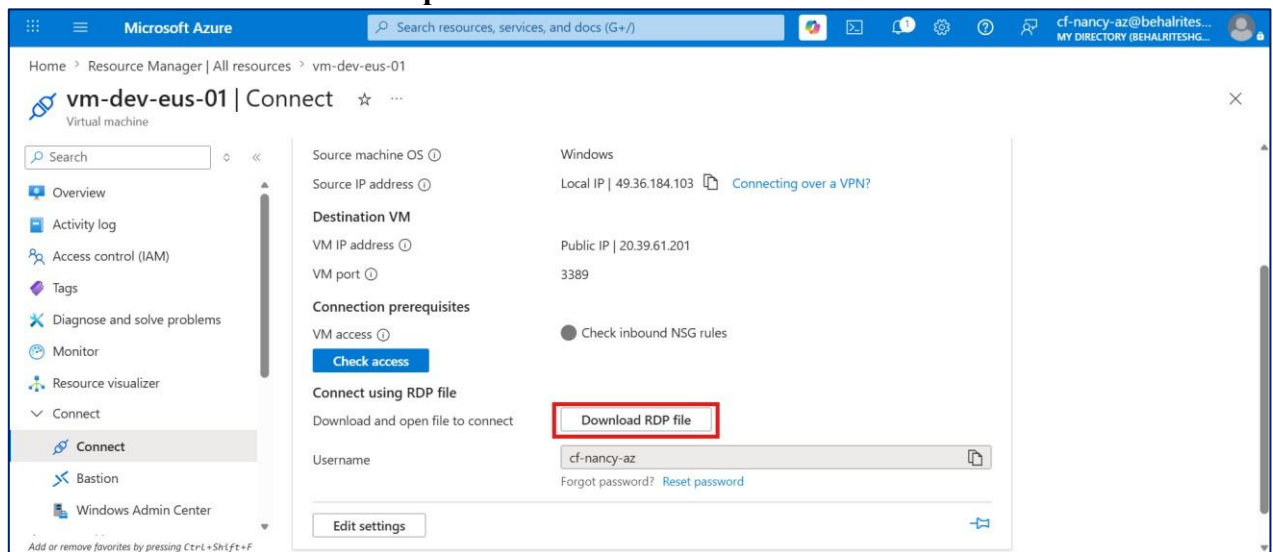


- Click Connect.



- Select RDP.
- Click Download RDP File.

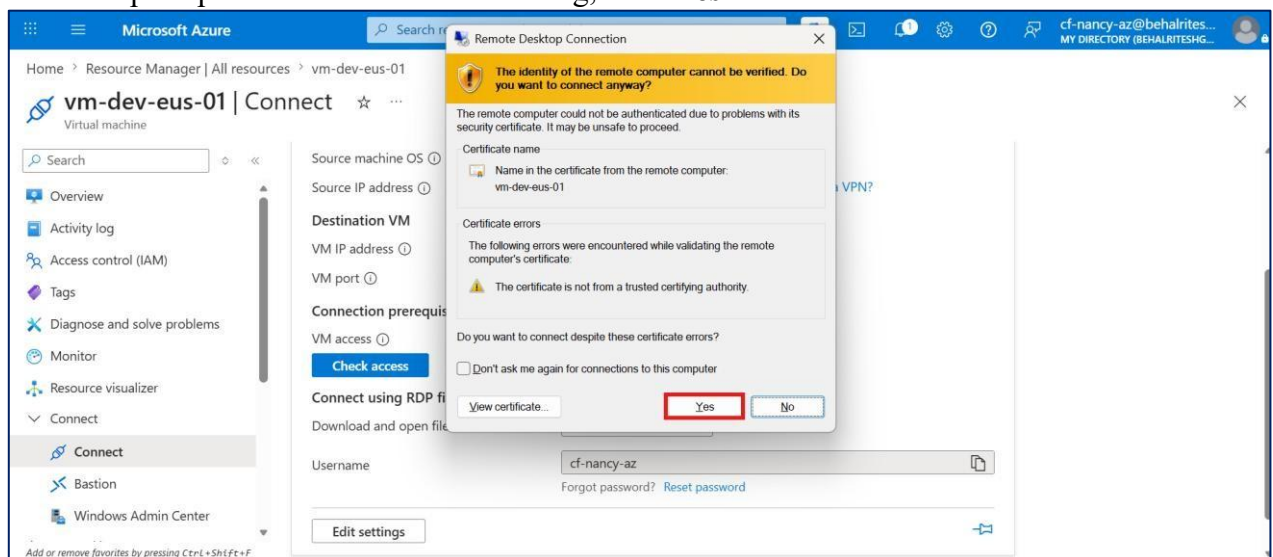
- Save the downloaded **.rdp** file.



- This step downloads the Remote Desktop configuration file required to connect to the virtual machine.

Step 14 – Connect to the Virtual Machine

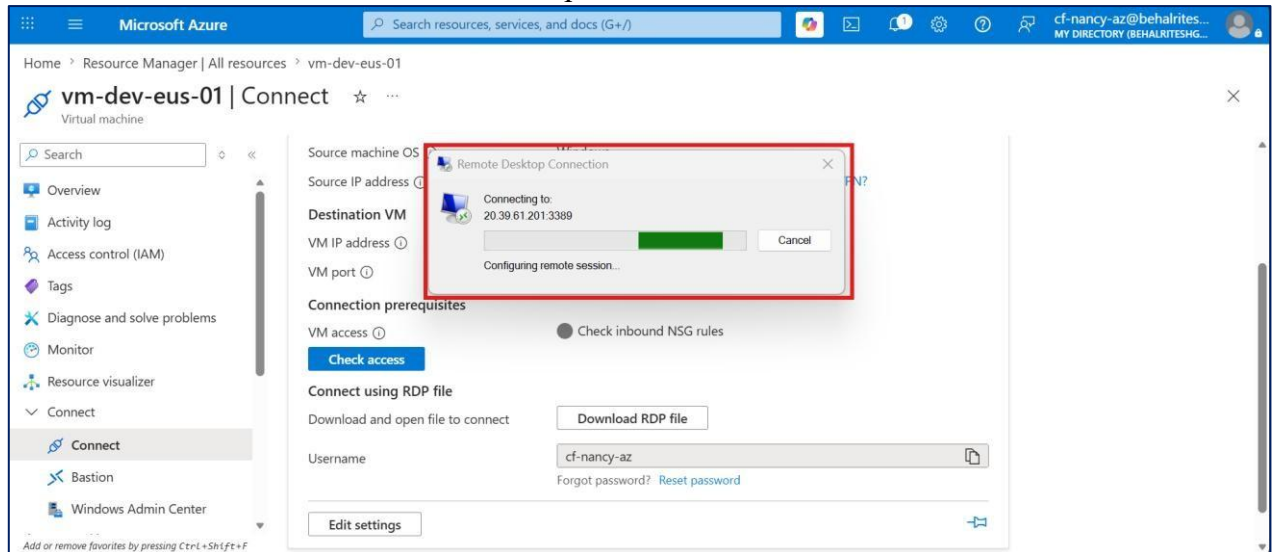
- Open the downloaded **.rdp** file.
- Click **Connect**.
- Enter the administrator username and password configured during deployment.
- If prompted with a certificate warning, click **Yes**.



- This step establishes a Remote Desktop connection to the Azure virtual machine.

Step 15 – Verify the Remote Desktop Session

- Wait for the Windows Server desktop to load.



- Verify that the Windows Server 2025 desktop is displayed.
- Confirm that the remote session is active and responsive.
- This step confirms successful Remote Desktop access to the Azure virtual machine.

Verification

- Successfully created a Resource Group.
- Successfully deployed a Windows Server 2025 virtual machine.
- Configured compute, networking, storage, and security settings.
- Verified the Azure resources created during deployment.
- Downloaded the Remote Desktop (RDP) connection file.
- Connected to the virtual machine using Remote Desktop.
- Verified successful access to the Windows Server 2025 desktop.